Noah Cosamano

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Objective: Seeking a Cybersecurity co-op using strong networking and programming skills in Python and Java to support secure systems design and threat detection and mitigation. Available May 2026 – August 2026

Education

Rochester Institute of Technology (RIT), Rochester, NY — Expected May 2028

Bachelor of Science, Cybersecurity

Related Courses: Programming for Information Security, Routing & Switching, Software Development and Problem Solving I and II, Introduction to Cybersecurity, Ethics in Computing, Discrete Mathematics for Computing, Reverse Engineering Fundamentals, Systems Administration

Skills

Programming Languages: Python, Java, C, C++, Assembly, SQL
Operating Systems: Kali Linux, MS Windows, iOS, and Cisco IOS
Tools & Platforms: PyTorch, Wireshark, Scapy, Nmap, Git, PuTTY, VS Code, Visual Studio

Projects

Network Intrusion Detection & Auto-Blocking System — March 2026 – Present

- Developed a real-time network monitoring tool in python to detect port scanning and other flaggable activities.
- Designed a multi-threaded processing system separating packet capture, parsing, and thread detection for speed.
- Automates threat mitigation by dynamically creating Windows Firewall rules to block malicious IP addresses.

Productivity & Habit Tracking System — February 2026 – Present

- Engineered a Python-based application that gamifies personal productivity using game-style progression.
- Implemented a relational SQL database to manage users, tasks, stats, and ranking data for scalability.

Buffer Overflow Exploit Simulation, Class Project — *August 2025 – December 2025*

- Analyzed a C-based file server and discovered a stack buffer overflow due to improper command parsing.
- Wrote Python scripts to send controlled input to server and find exact offset needed to overwrite instruction pointer.
- Used x32dbg to inspect CPU registers and take control of program execution.
- Redirected execution flow using a jmp esp instruction and displayed how malicious code can be executed from stack.
- Implemented NOP sled and custom shellcode to simulate a reverse TCP connection and remote command execution.
- Researched and documented common defenses such as ASLR, stack canaries, PIE, and Control Flow Guard, and explained how they reduce exploitability.
- Authored a formal vulnerability report detailing discovery, exploitation, and impact of a buffer overflow vulnerability.

Network Packet Crafter, Personal Project — August 2025 – September 2025

- Developed tool in Python using VS Code IDE, and Git for version control to craft and send custom packets at the command line with customization through user-defined source and destination IP/MAC/port/payload values.
- Supported TCP, UDP, ARP, ICMP, and ICMPv6 protocols; enabled dynamic packet routing based on user input.
- Logged packet metadata to an SQL database with hashed IPv4, MAC, and payload for added security.
- Validated and debugged packets through Wireshark during testing.

Work Experience

McDonalds' June 2025 – Present Crew Member Fairmount, NY

- Delivered fast, high-quality customer service while managing a wide range of responsibilities under pressure.
- Collaborated with team members to meet standards to ensure consistent customer satisfaction.